

# Kyler Jeong

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## EDUCATION

### University of California, Berkeley

GPA: 3.65

*B.S. Mechanical Engineering*

*Aug 2024 – May 2028*

**Relevant Coursework:** Introduction to Python and MATLAB, Physics Mechanics, Physics E&M, 3D Modeling for Design, Electronics for IoT, Design Methodology, Statistics and Data Science for Engineers, Thermodynamics, Solid Mechanics, Fluid Mechanics, Manufacturing and Design Communication, Introduction to Product Development

## TECHNICAL SKILLS

**Design:** SolidWorks, AutoCAD, Fusion360, Onshape, 3D CAD, Geometric Design and Tolerancing (GD&T), Design for Manufacturing (DFM), Engineering Drawings

**Engineering and Analysis:** Finite Element Analysis (FEA), Electrical Engineering, Arduino, ESP32, Soldering, MATLAB, Python, MicroPython, C++

**Manufacturing and Prototyping:** Additive Manufacturing (3D Printing), Laser Cutting, CNC Machining, Rapid Prototyping, Hands-on Machining, Computer Aided Manufacturing (CAM)

## EXPERIENCE

### Development Engineering Co-op

Aug 2026 – Dec 2026

*Bosch*

*Charleston, SC*

- Supporting design and validation of low and high-pressure gasoline engine components, performing teardown analysis on field-return and development-tested parts to identify failure locations and surface conditions

### Research Trainee Intern

Jun 2026 – Aug 2026

*Integrative Movement Sciences Institute*

*Irvine, CA*

- Engineered a robotic foot design from biomechanical research on guinea fowl morphology at the UCI Neuromechanics Lab, building a force-plate rig to benchmark ground reaction forces against existing models

### Executive Treasurer

Aug 2025 – Present

*Combat Robotics at Berkeley*

*Berkeley, CA*

- Co-designed and manufactured multiple 1lb and 3lb combat robots, optimizing component layouts for high-impact durability and machining custom chassis components
- Responsible for making executive decisions regarding the club and coordinating with Program Leads to create and manage the budget for the academic year, allocating over \$20,000 in funds to over 30 technical projects

## PROJECTS

### Avian-Inspired Robotic Foot | 3D CAD, Experimental Testing, Rapid Prototyping

Jun 2026 – Present

- Designed and 3D printed over 10 prototype iterations of an avian-inspired foot design, testing it with a custom configured BirdBot leg design
- Utilized MATLAB to analyze and quantify test data, comparing ground reaction forces from different foot models

### Beetleweight Beater Bar Combat Robot | CNC Machining, DFM, FEA

Dec 2025 – Present

- Engineered a 3lb beetleweight combat robot from concept to physical prototype, implementing GD&T and DFM principles to streamline CNC machining of the weapon assembly
- Developed a second-generation iteration to resolve failure points identified during testing of the initial prototype

### SunGlow Plant Pot & Lamp | GD&T, Additive Manufacturing, Design Methodology

Jan 2026 – May 2026

- Designed and manufactured an integrated LED planter utilizing a passive self-watering reservoir and refractive optics to provide both ambient lighting mimicking natural sunlight and functional plant growth
- Created a working prototype that successfully achieved Mie/Rayleigh scattering, mimicking the tone and refraction of natural sunlight

### AI Integrated Phone Lock Box | Mechatronics, Product Design, Rapid Prototyping

Jan 2026 – May 2026

- Led a 6-person team to prototype an IoT smartphone lockbox integrating an ESP32, OLED display, and mechanical locking mechanism programmed via Python and C++
- Developed a custom enclosure in Onshape utilizing laser cutting and additive manufacturing, optimizing the design using customer feedback, market research, and other design methodology tools